



CUSTOMER PERSONALITY PREDICTION

● TO BOOST MARKETING CAMPAIGN

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GREETINGS!

ABOUT ME

A Career Shifter with hands-on experience in Mortgage Product team. Currently transitioning into an Analyst role by joining Data Science programs by Rakamin and Digitalent to developed practical skills in Python and SQL to process, analyze, and visualize large datasets, uncover patterns, and derive meaningful insights.

[Link to Workbook](#)

DATASET SUMMARY

Data Source

marketing_campaign_data

Total Column

29 Columns

Total Row

2208

Type Data

1 ID

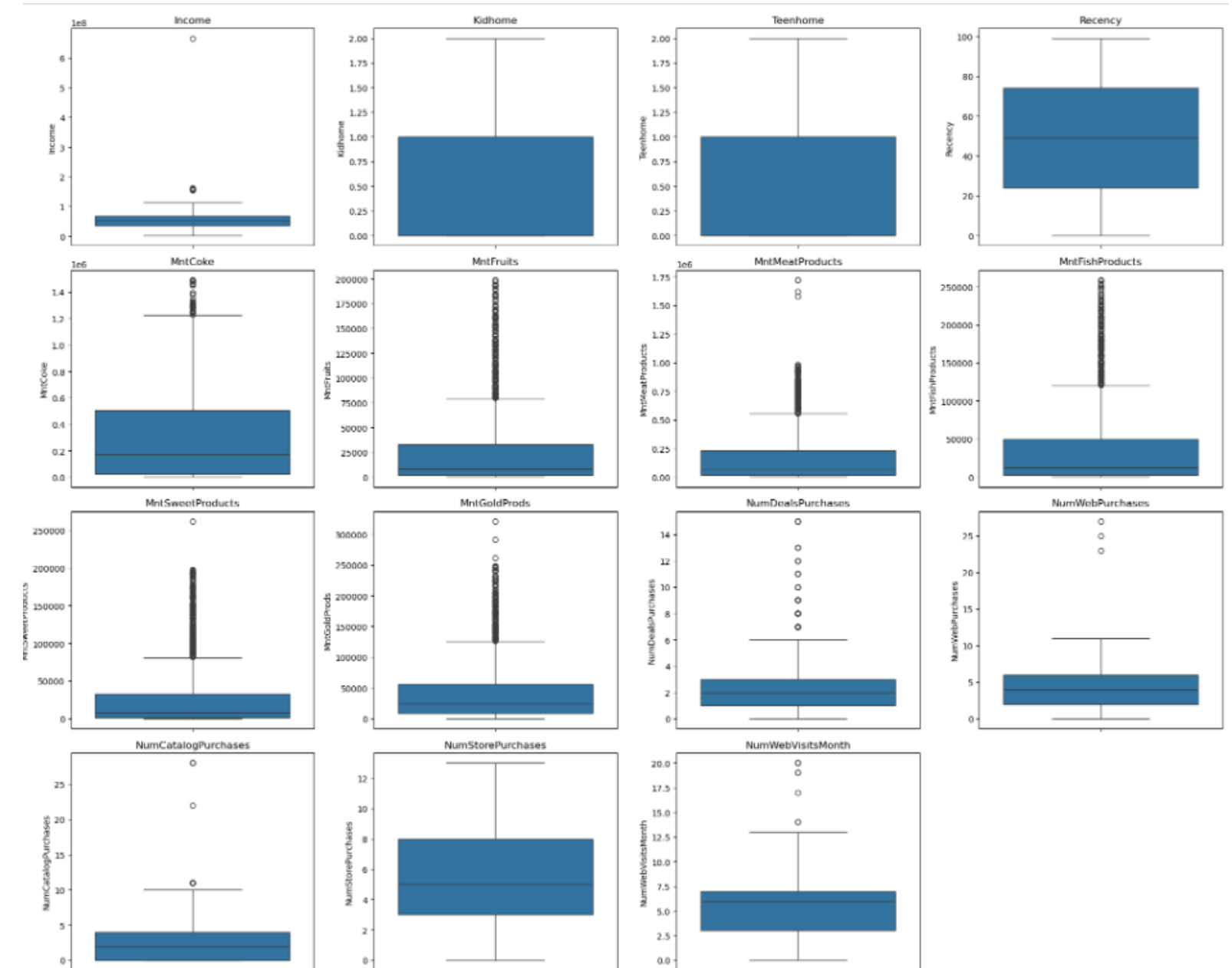
16 Numericals

11 Categoricals

Column	Data Type	Example Value	Range / Categories
Unnamed: 0	int64	0	0 – 2,239
ID	int64	5524	0 – 11,191
<u>Year Birth</u>	int64	1957	1,893 – 1,996
Education	object	S1	S1, S3, S2, SMA, D3
<u>Marital Status</u>	object	<u>Lajang</u>	<u>Lajang, Bertunangan, Menikah, Cerai, Janda</u>
Income	float64	58138000	1,730,000 – 666,666,000
<u>Kidhome</u>	int64	0	0 – 2
Teenhome	int64	0	0 – 2
Dt_Customer	object	4/9/2012	Date
Recency	int64	58	0 – 99
MntCoke	int64	635000	0 – 1,493,000
MntFruits	int64	88000	0 – 199,000
MntMeatProducts	int64	546000	0 – 1,725,000
MntFishProducts	int64	172000	0 – 259,000
MntSweetProducts	int64	88000	0 – 263,000
MntGoldProds	int64	88000	0 – 362,000
NumDealsPurchases	int64	3	0 – 15
NumWebPurchases	int64	8	0 – 27
NumCatalogPurchases	int64	10	0 – 28
NumStorePurchases	int64	4	0 – 13
NumWebVisitsMonth	int64	7	0 – 20
AcceptedCmp3	int64	0	0 – 1
AcceptedCmp4	int64	0	0 – 1
AcceptedCmp5	int64	0	0 – 1
AcceptedCmp1	int64	0	0 – 1
AcceptedCmp2	int64	0	0 – 1
Complain	int64	0	0 – 1
Z_CostContact	int64	3	3 – 3
Z_Revenue	int64	11	11 – 11
Response	int64	1	0 – 1

EXPLORATORY DATA

Aspect	Result	Explanation
Missing Values	26 missing values	All columns have complete data, with no entries containing 'none'
Duplicates	0 duplicate	There is no duplicate data
Outliers	4 Outliers	Total 4 outlier identified. 3 from Year Born and 1 from Income.
Invalid Data	3 Invalid data	There's 3 data where customer born in 1800s and 1 data where the income extremely high while spending & purchases low.
Format Difference	1 column	All data has a uniform format, with no differences in letter casing



FEATURE ENGINEERING

To further analyze the customer profile and behavior, these features are engineered

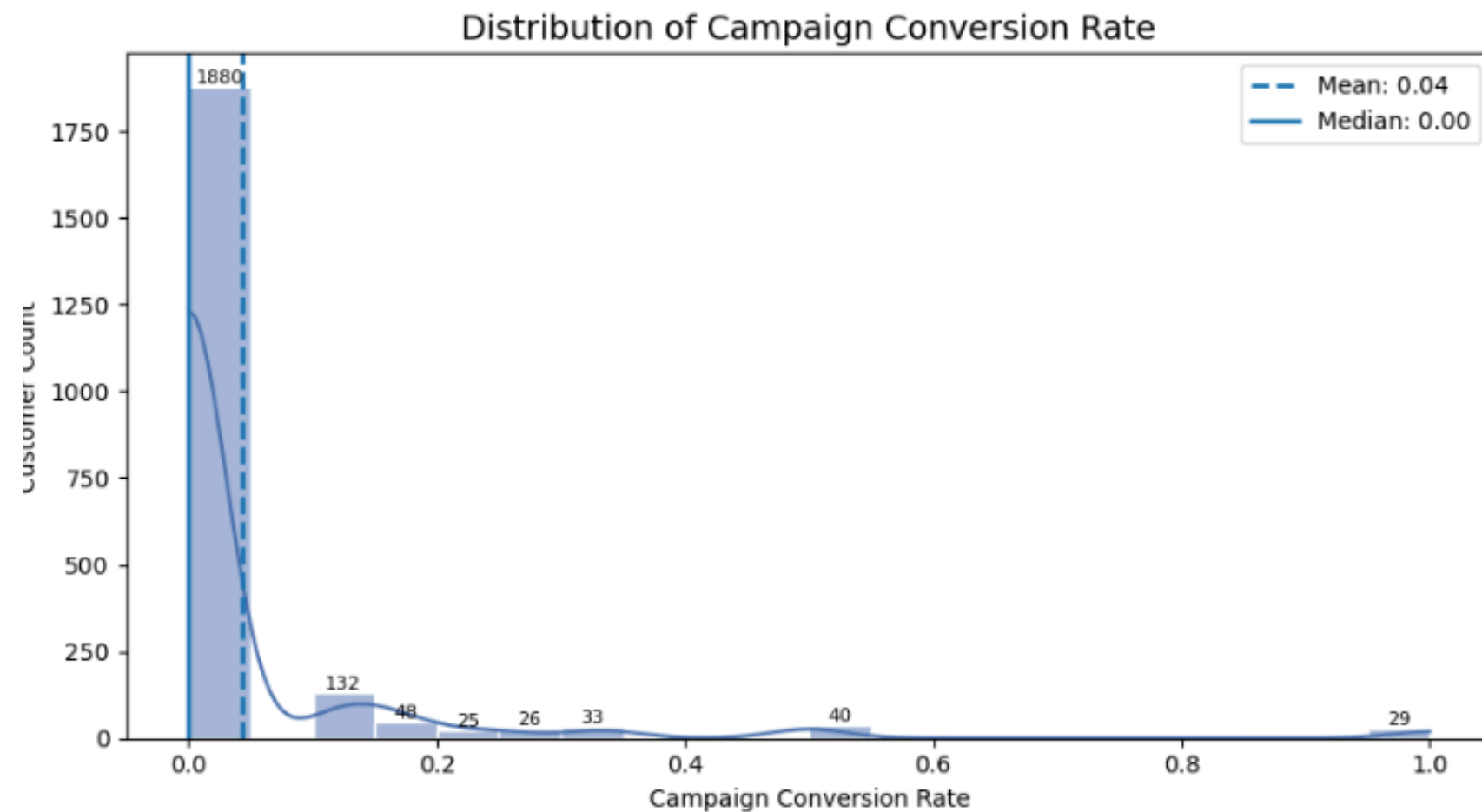
Features	Formula	Function/Purpose
Age	today.year - Year_Birth	Measure the customer's age based on their year of birth
Total Child Home	Kidhome + Teenhome	Summarized total children at home
Total Amount Spending	MntCoke + MntFruits + MntMeatProducts + MntFishProducts + MntSweetProducts + MntGoldProds	Summarized how much the customers has spend from buying the products from the company
Tenure	today.days - Dt_Customers	New vs loyal customers behave very differently.
Total Purchases	NumWebPurchases + NumCatalogPurchases + NumStorePurchases	Summarized how often customer shop from the company

Features	Formula	Function/Purpose
Channel Ratio	Web_Ratio = NumWebPurchases / Total_Purchases Store_Ratio = NumStorePurchases / Total_Purchases	To describe the customer's channel preference
Deal Ratio	NumDealsPurchases / Total_Purchases	Identifies price-driven vs value-driven customers.
Campaign Acceptance Rate	AcceptedCmp1+...+AcceptedCmp5)/5	Summarized all historical campaign result
Campaign Conversion Rate	NumWebPurchases + NumCatalogPurchases + NumStorePurchases	Summarized how often customer shop from the company

The engineered **features will be segmented into Customer Profile and Customer Behavior clusters** to identify customer personality patterns and support more effective marketing strategies.

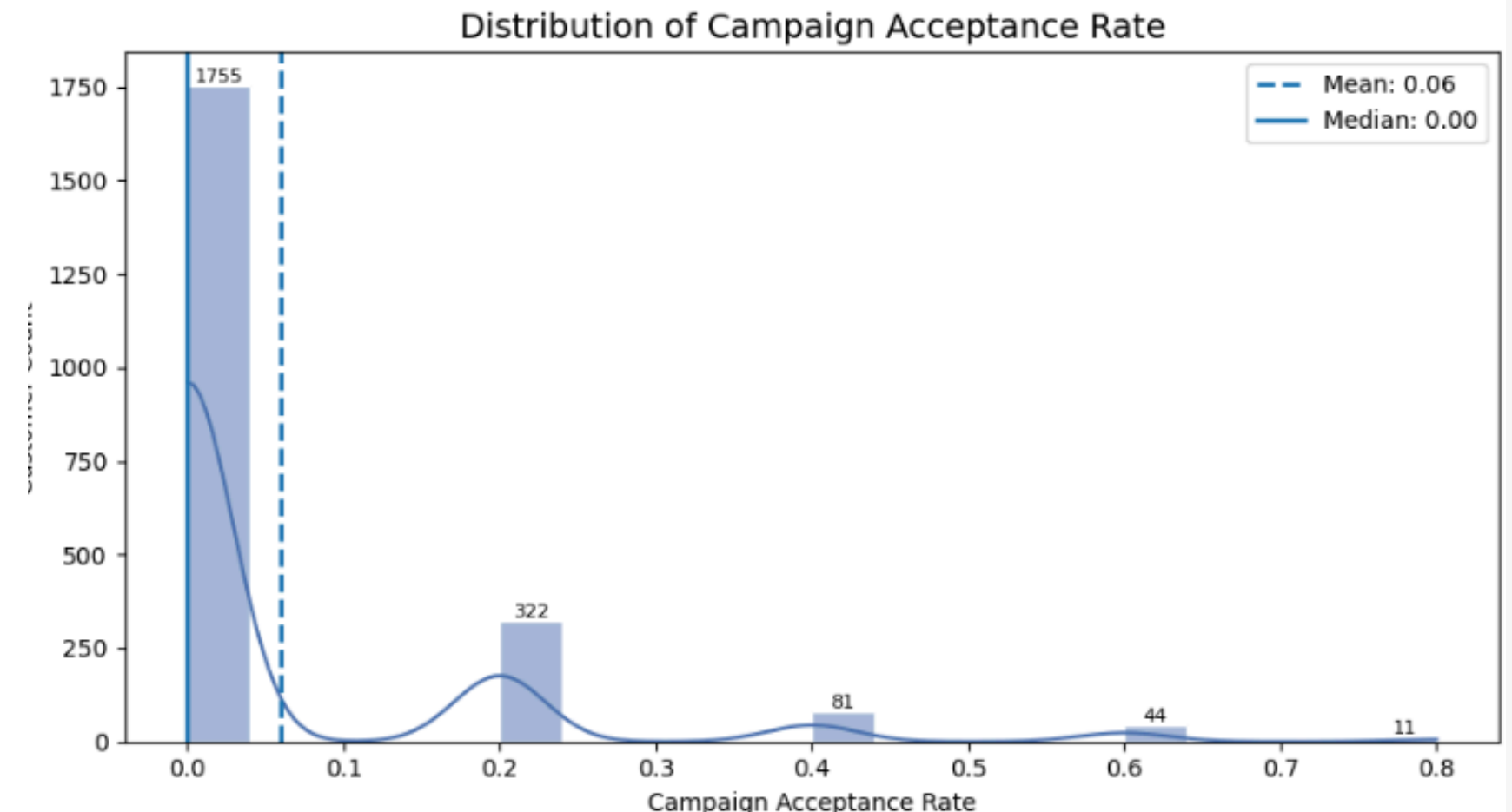
PROBLEMS

WHAT SHOWN ON THE DATASET?



Web Campaign Conversion Rate

Overall, most of **customers** in the database **have low Web Campaign Conversion Rate**. Where the **average Campaign Conversion Rate only 4%**



Campaign Acceptance Rate

Majority of customer's too have low Campaign Acceptance Rate. Out of total 2213 customers **only 458 (21%)** that ever **respond to any campaigns.**

A photograph of two men in a meeting. One man, wearing a black leather jacket, is holding a smartphone. The other man, wearing a white jacket, is writing in a notebook. They are sitting at a table with some papers and a coffee cup. The background is slightly blurred.

WHAT CAN BE DONE?

ABOUT OUR GOALS AND OBJECTIVE

GOALS

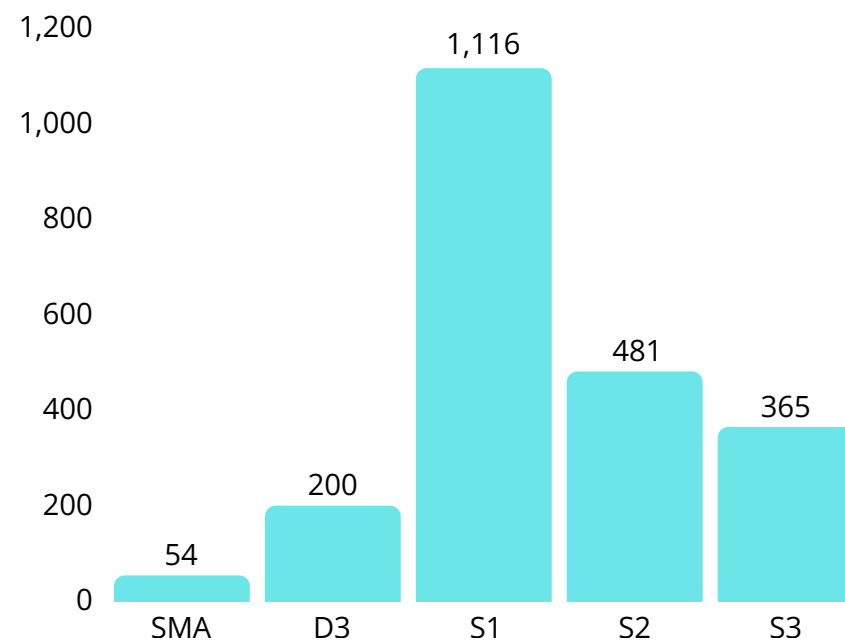
Develop marketing strategies that tailored to the customers personality to boost marketing campaign, increases revenue contribution and optimizes campaign targeting efficiency.

OBJECTIVE

- Identify high-value customer personas
- Optimize campaign targeting using behavior clusters
- Increase marketing ROI through precision segmentation
- Develop actionable customer engagement strategies

CUSTOMER PROFILE

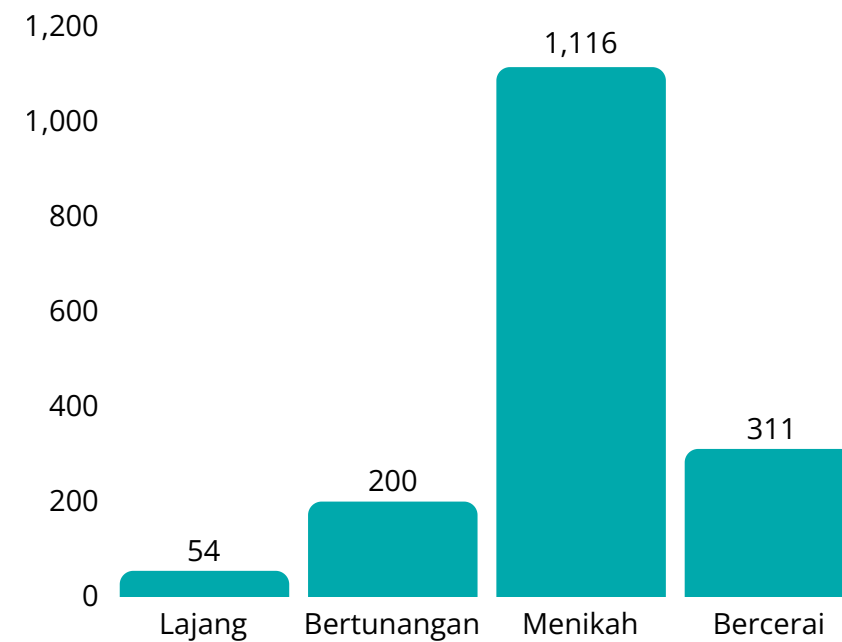
Education



Education

Based on education, most of the company customers have rather high education degree.

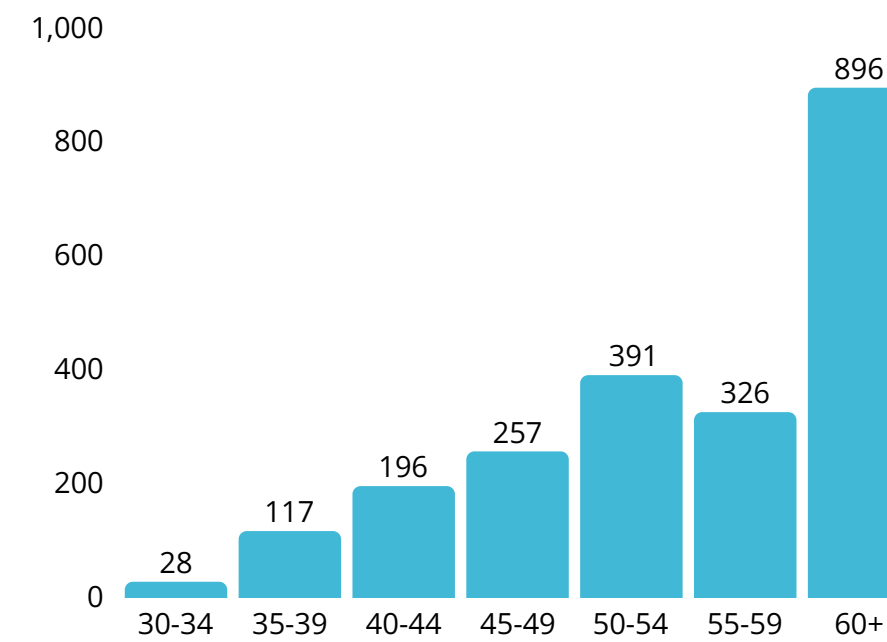
Marital Status



Data-Driven Marketing

Most of our customers are married.

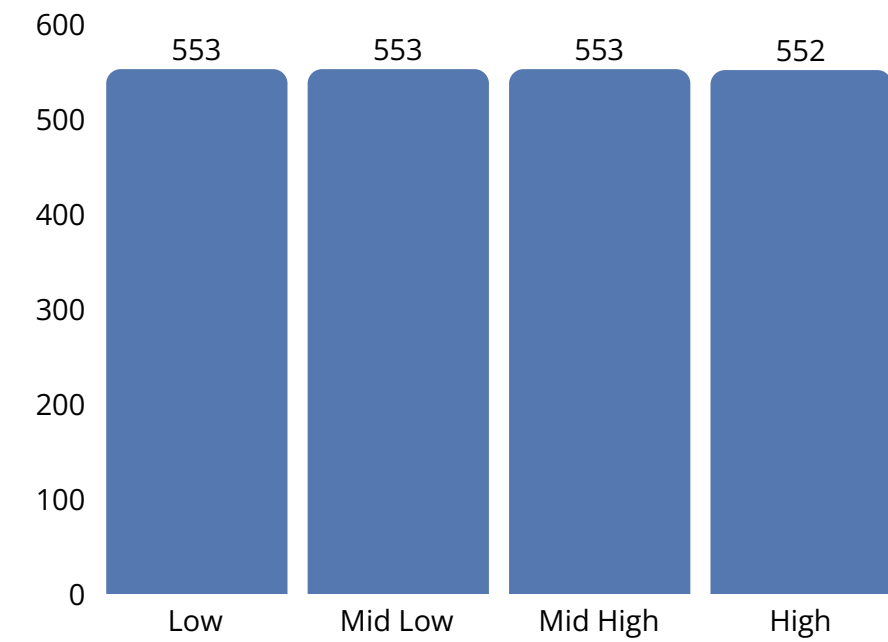
Age Range



Age Range

The company business are dominated by customers with 60+ age range. In the future the company may need to attract more of younger generation.

Income Group

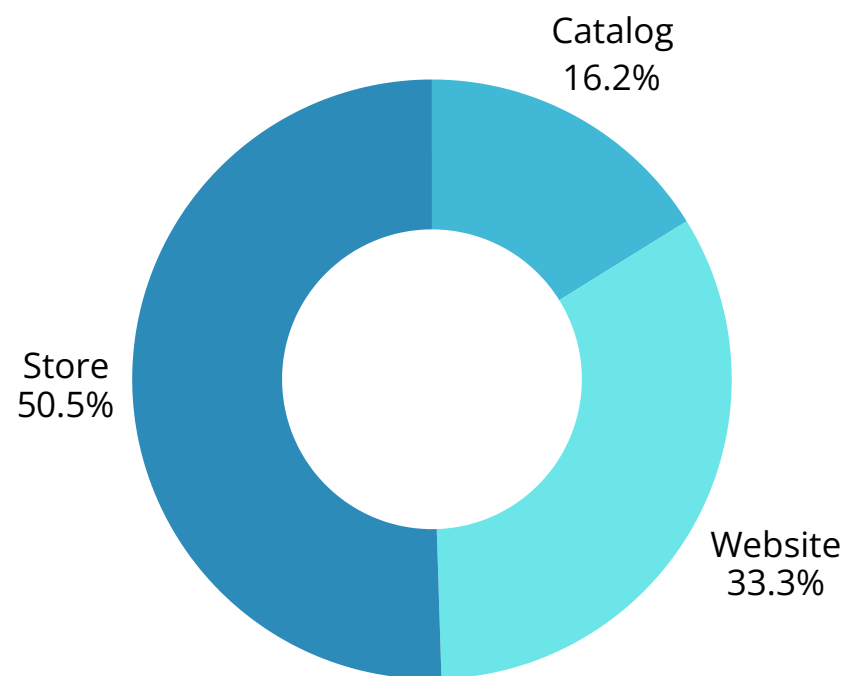


Income Group

The customers distributed rather evenly based on the Income Group with numbers of each group rather identical

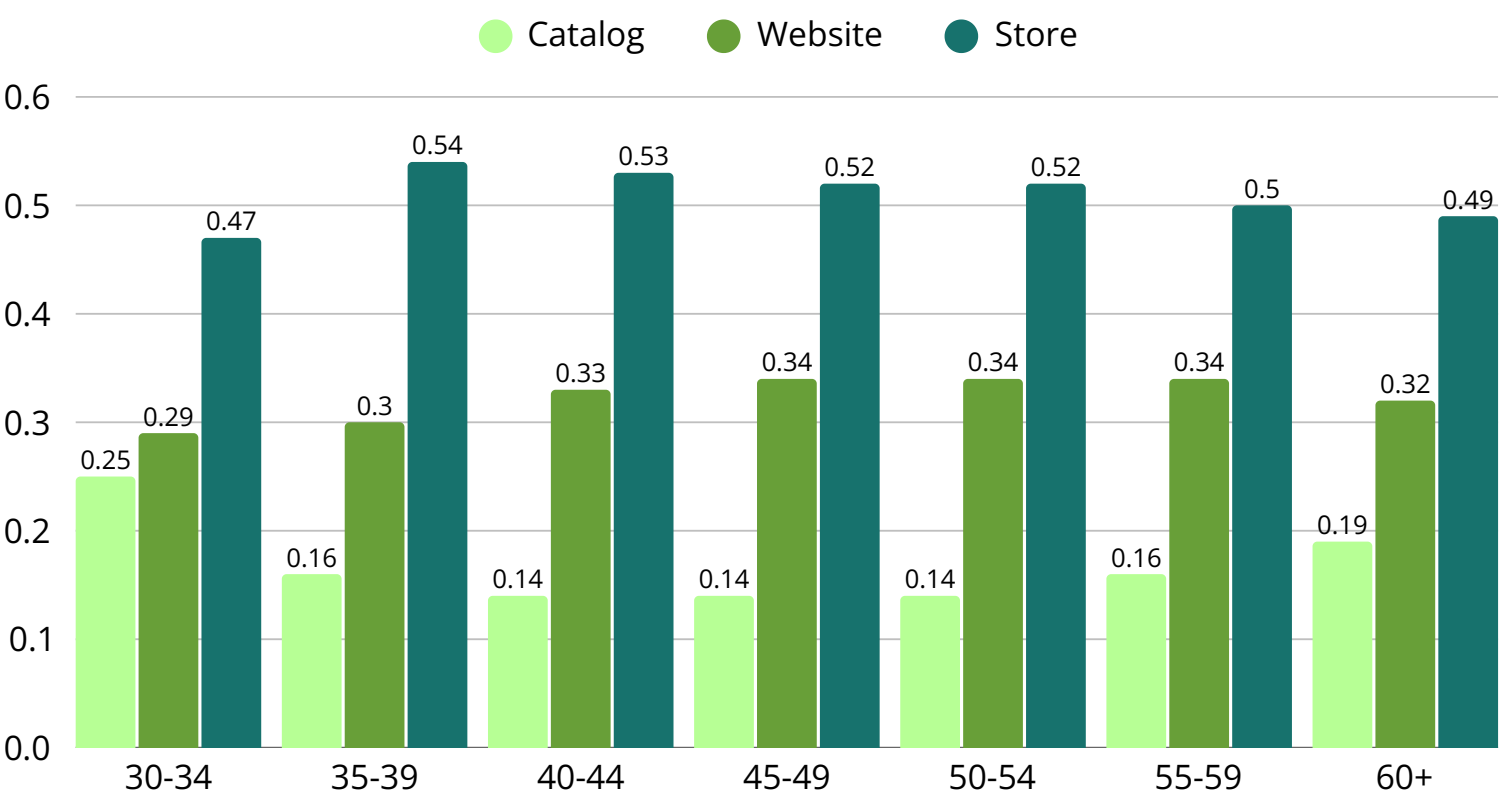
CUSTOMER'S BEHAVIOR BY CHANNELS

Channels Distribution

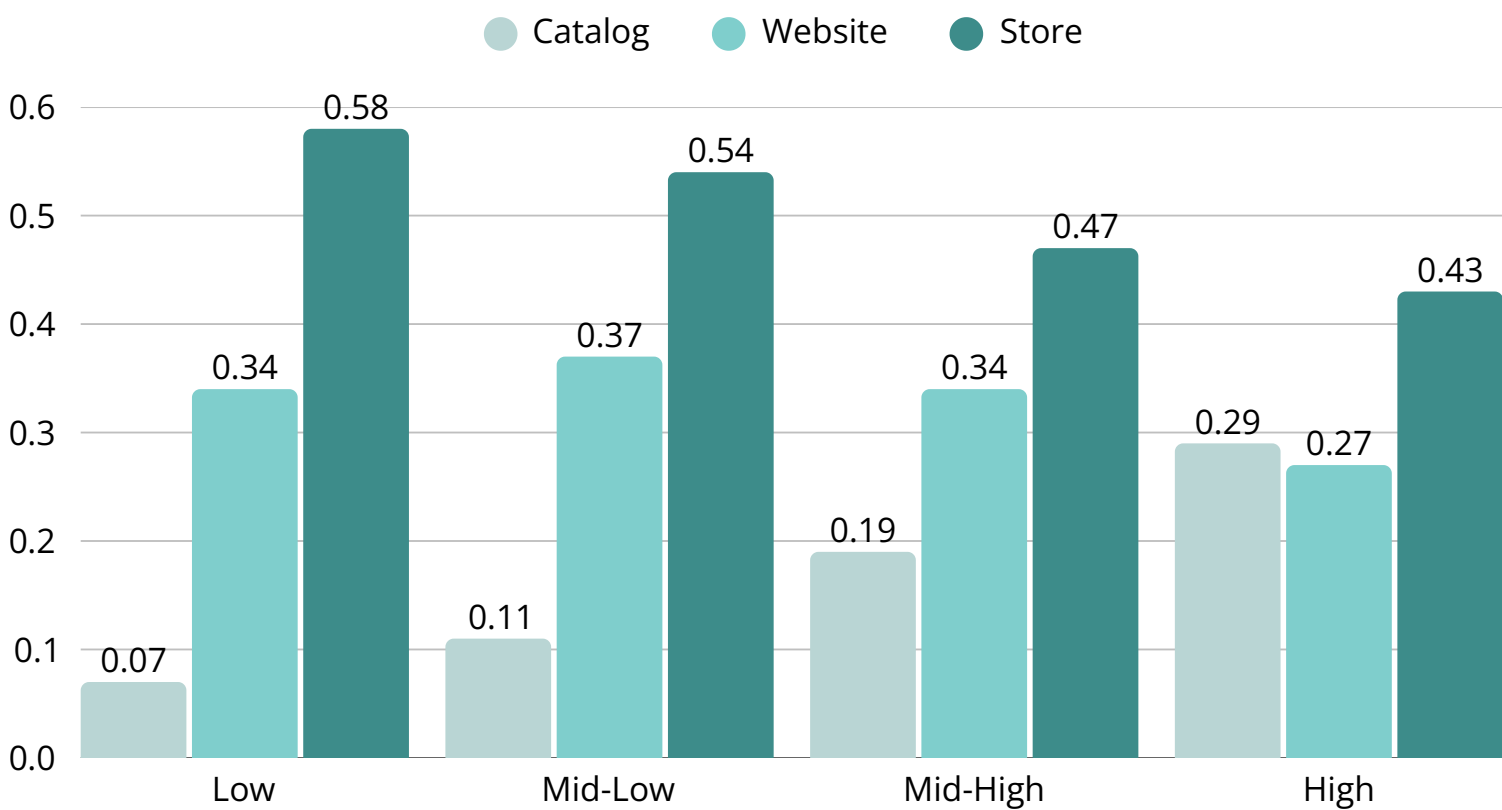


- Physical stores remain the primary touchpoint, accounting for 50.5% of total distribution across all demographics.
- Website engagement remains remarkably stable at approximately 33% regardless of age, indicating a platform with broad, cross-generational appeal.
- High income group is the most diverse in its channel usage, showing a unique 29% engagement rate with catalog.

Channel Ratio by Age

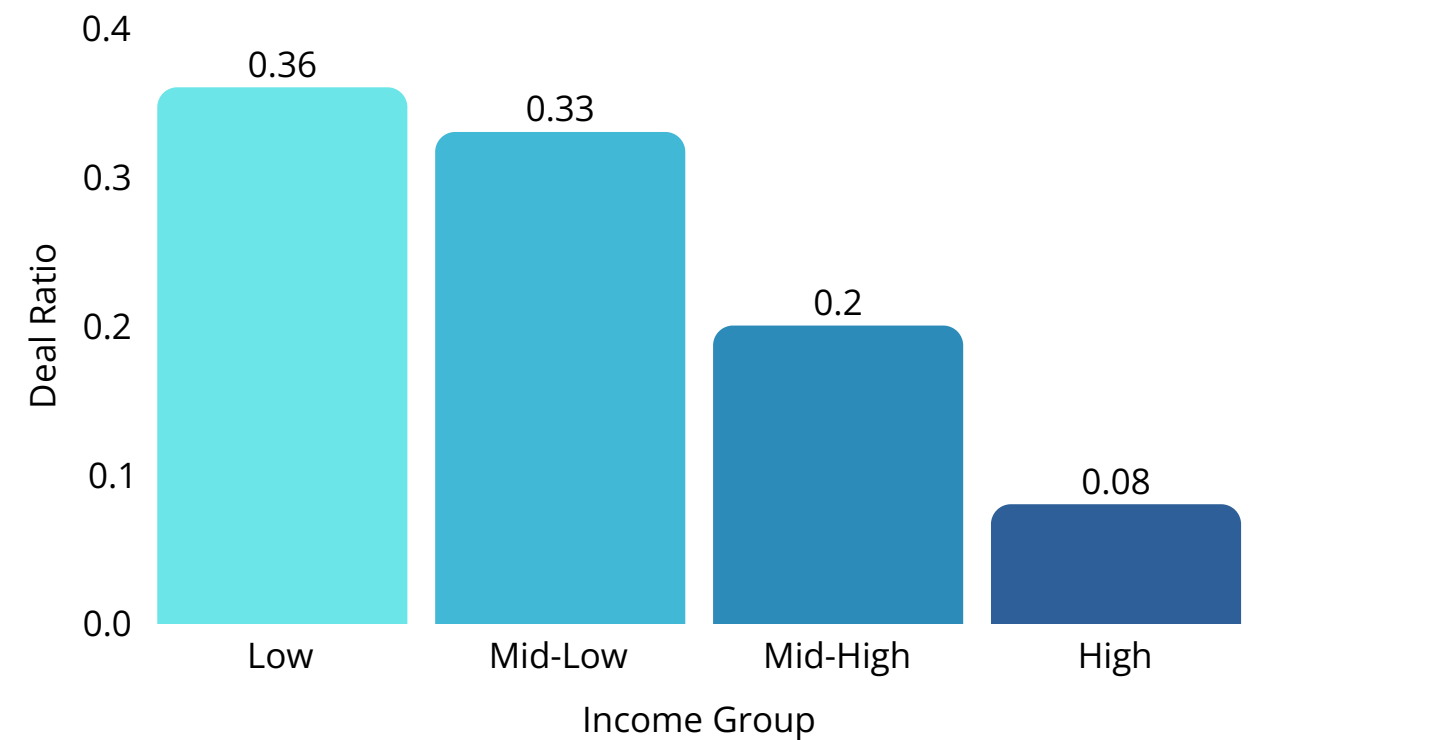


Channel Ratio by Income



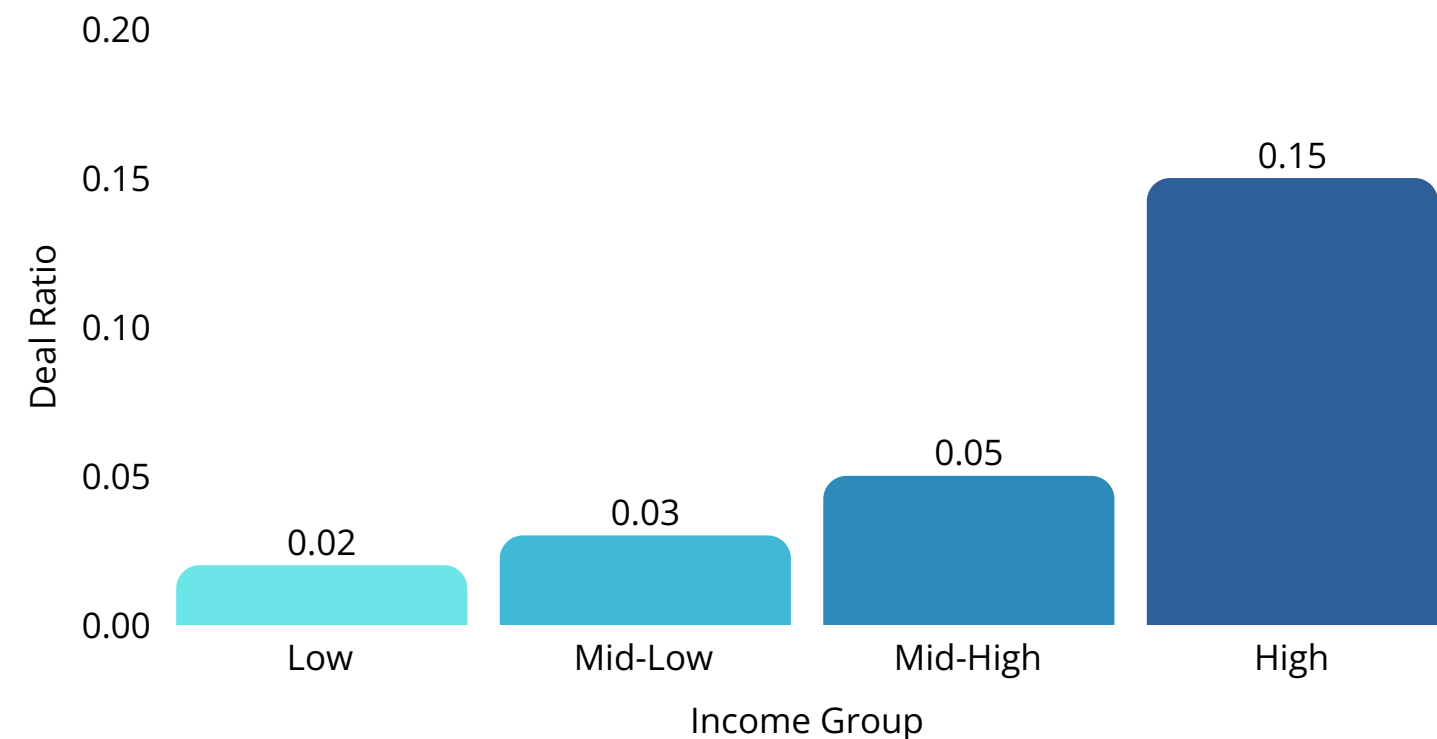
CUSTOMER'S BEHAVIOR HIGHLY AFFECTED BY INCOME GROUP

Deal Ratio by Income Group

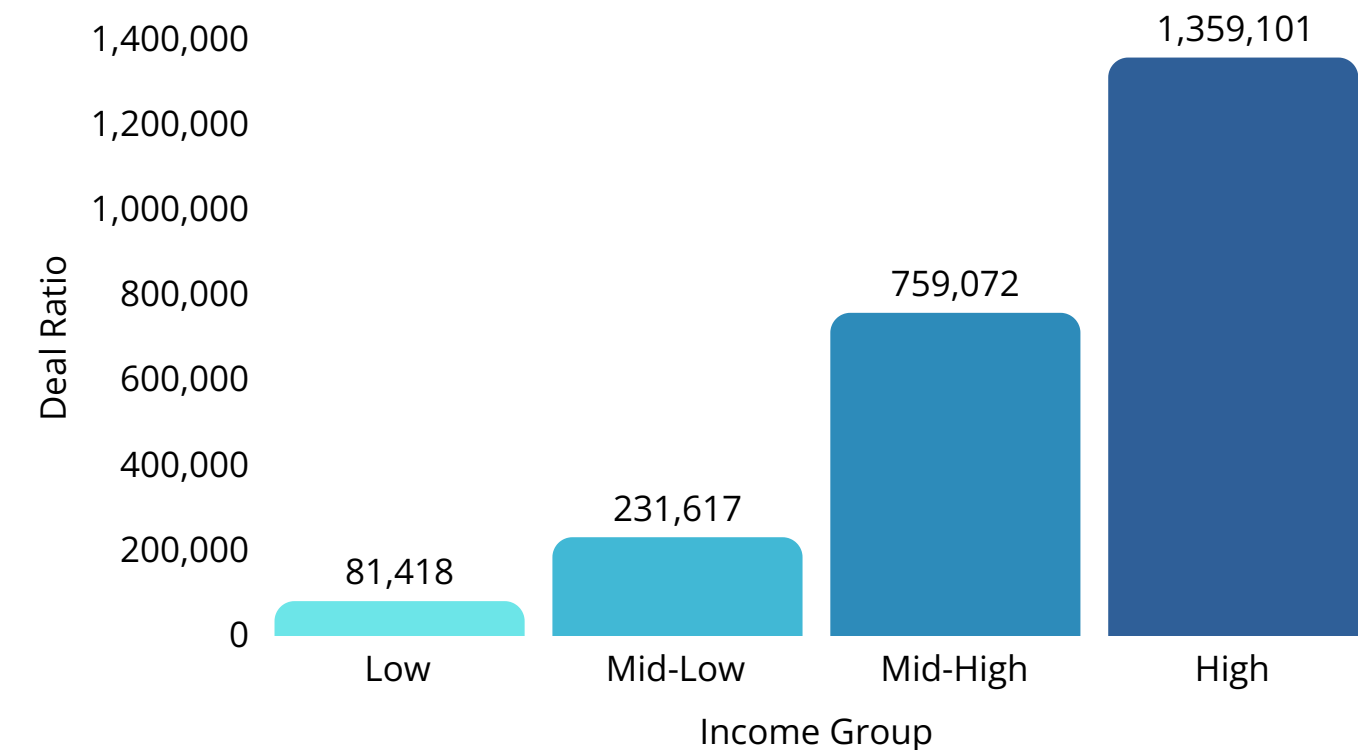


- Low-income groups are the most "deal-hungry", but they contribute the least to total revenue.
- While high-income groups rarely wait for a deal , they are likely to accept a campaign offer and spend over more than the low-income groups.
- Marketing campaigns are significantly more effective as income rises.

Campaign Acceptance Rate by Income Group



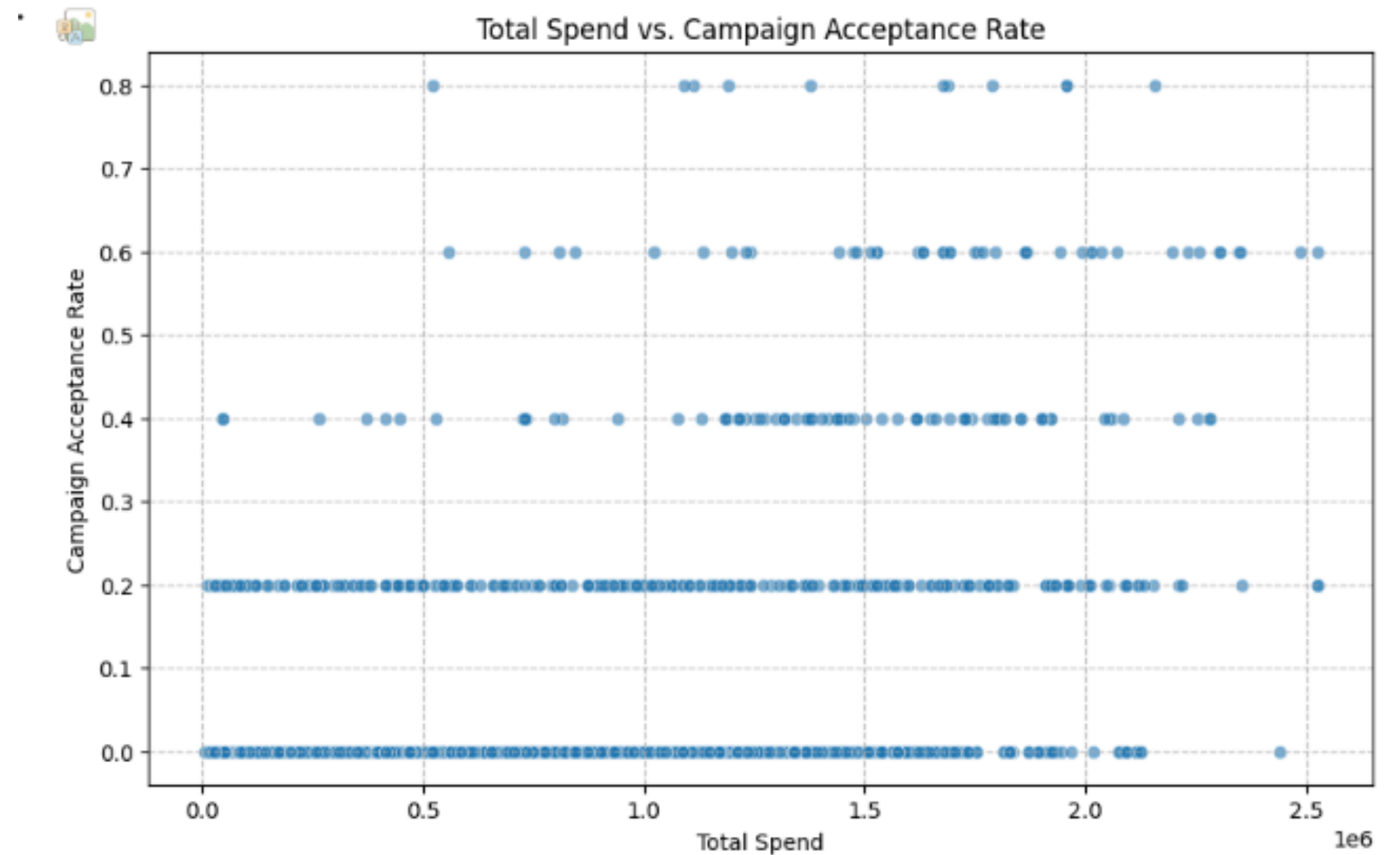
Total Spending by Income Group



CAMPAIGN

WHAT HAPPEND?

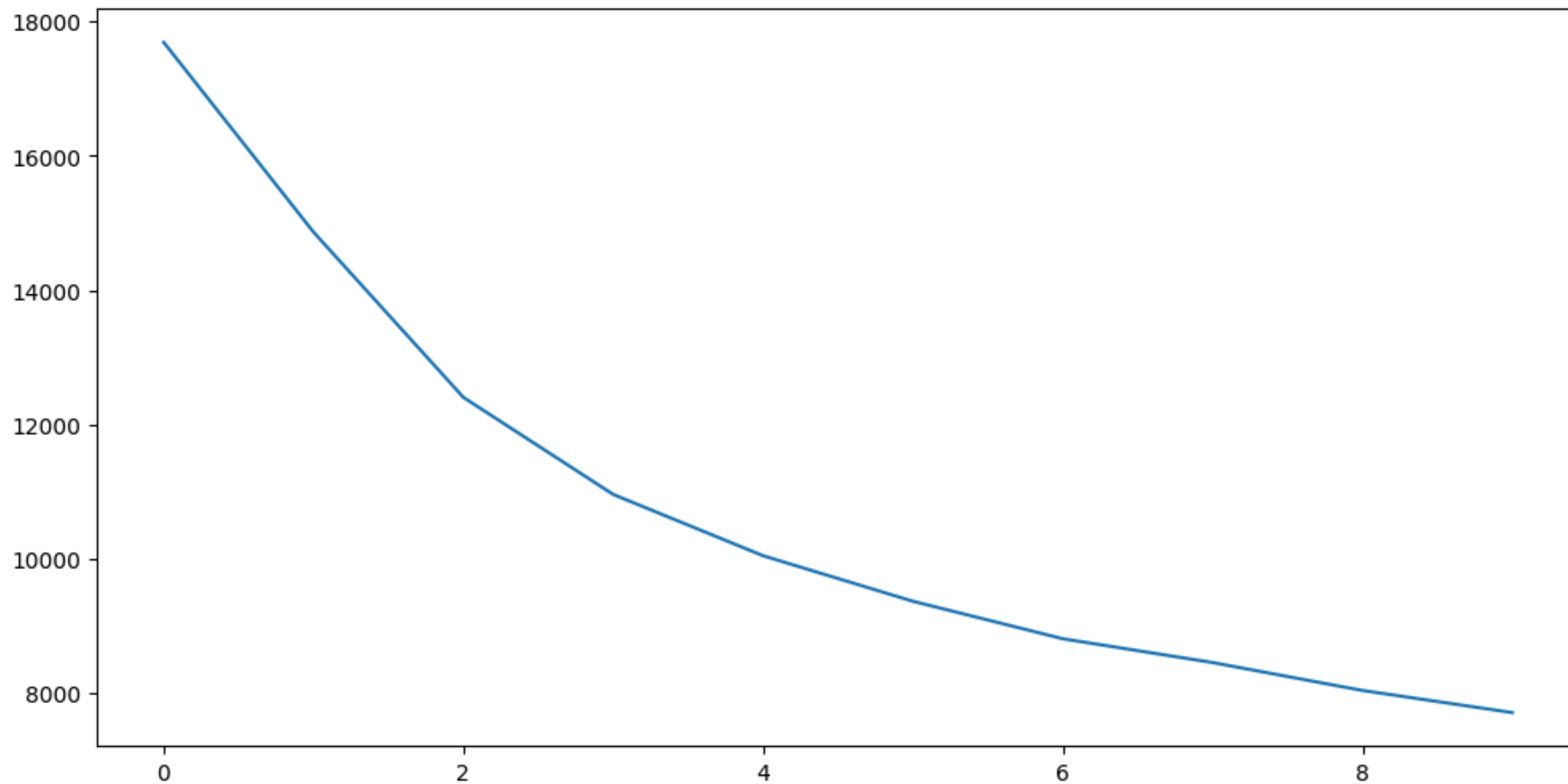
Campaign acceptance is spread across all spending tiers, meaning revenue is not a strong predictor of engagement. This unlocks a clear opportunity: **Move from value-based targeting to behavior-driven segmentation to drive higher campaign efficiency and ROI.**



- Customer spending level does not strongly influence campaign acceptance.
- Most customers cluster at low acceptance levels (0–20%)
- Campaign effectiveness not depend only to spending habit but also other factors such as behavior.



CLUSTERING - PROFILE



```
*** k=2, Silhouette Score=0.1605
    k=3, Silhouette Score=0.2540
    k=4, Silhouette Score=0.2006
    k=5, Silhouette Score=0.1953
    k=6, Silhouette Score=0.1840
    k=7, Silhouette Score=0.1917
    k=8, Silhouette Score=0.1873
    k=9, Silhouette Score=0.1924
    k=10, Silhouette Score=0.1947
```

Result

Based on the Elbow Method and silhouette score, the most optimal number of cluster is 3. Based on these result the cluster then divided into 3 segment: Mature Established Families, Mid-Age Balanced Segment, Younger & Early-Stage Families

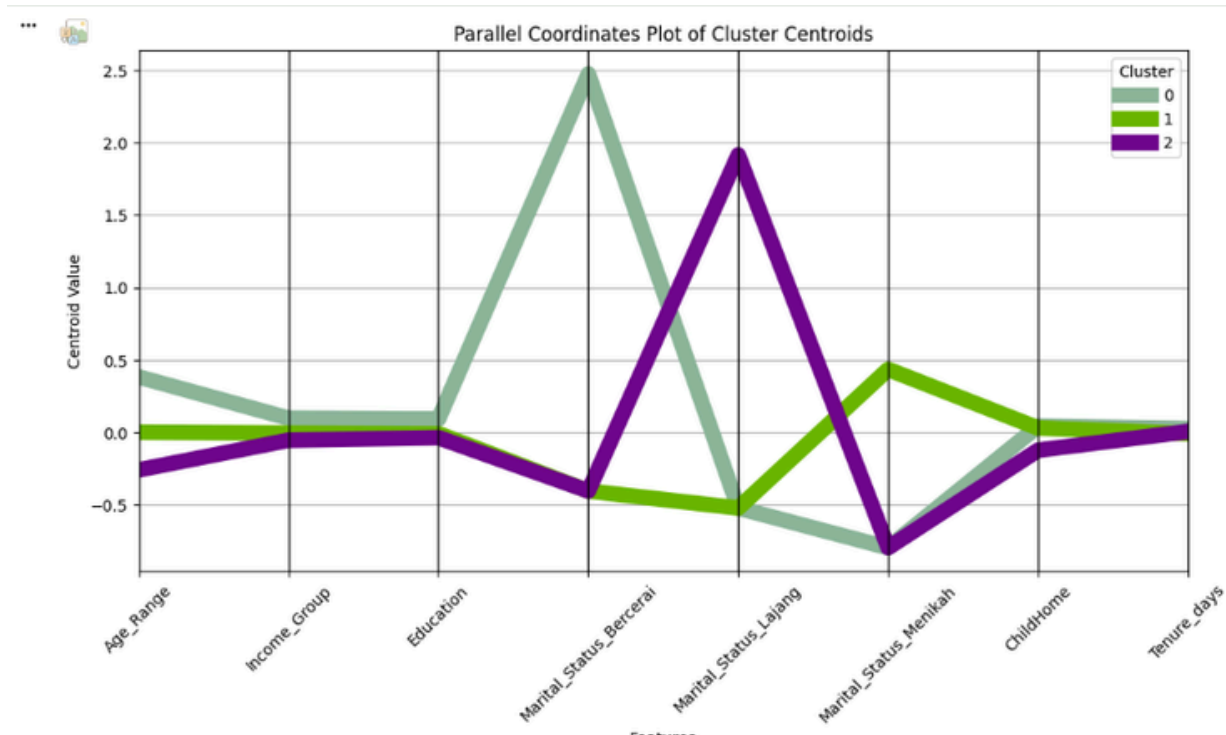
CLUSTERING

--- PROFILE

	Education	ChildHome	Tenure_days
Profile_Cluster			
0	2.554839	0.980645	4620.148387
1	2.452381	0.971289	4612.918067
2	2.424947	0.856237	4615.951374

Age_Range	0	1	2	3	4	5	6
Profile_Cluster							
0	0.000000	0.022581	0.012903	0.070968	0.167742	0.200000	0.525806
1	0.007703	0.042717	0.097339	0.125350	0.186975	0.142157	0.397759
2	0.035941	0.103594	0.112051	0.118393	0.152220	0.128964	0.348837

Income_Group	0	1	2	3
Profile_Cluster				
0	0.193548	0.258065	0.293548	0.254839
1	0.253501	0.243697	0.257003	0.245798
2	0.276956	0.264271	0.200846	0.257928



00 Mature Established Families

Highest percentage in oldest (60+) age group, dominant in Mid - High Income group, higher education average, Long tenure

- Stable, mature, financially capable customers

01 Mid-Age Balanced Segment

Dominated by customers in 40-59 age group. Income distributed evenly and have long tenure

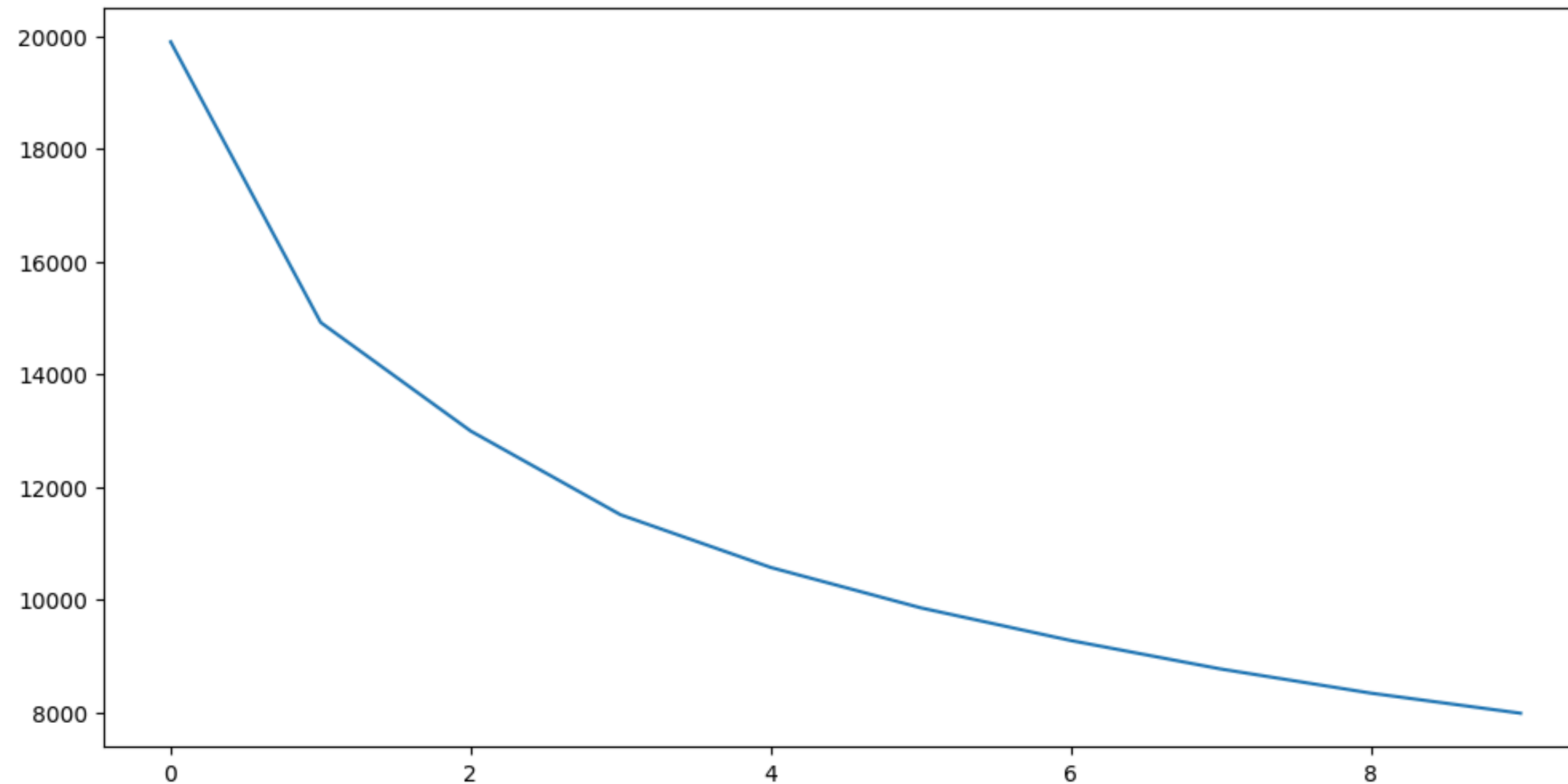
- Stable middle-market base.

02 Younger & Early-Stage Families

Higher propotions in younger customers and lower income group than other clusters also have long tenure

- Younger but economically mixed segment.

CLUSTERING - BEHAVIOUR



```
k=2, Silhouette Score=0.2556  
k=3, Silhouette Score=0.2289  
k=4, Silhouette Score=0.1905  
k=5, Silhouette Score=0.1877  
k=6, Silhouette Score=0.1926  
k=7, Silhouette Score=0.1736  
k=8, Silhouette Score=0.1657  
k=9, Silhouette Score=0.1586  
k=10, Silhouette Score=0.1569
```

Result

Based on the Elbow Method and silhouette score, the most optimal number of cluster is 2. However we decide to use 4 cluster so we can make 4 behaviour segment so the profile x behaviour segmentation will be more diverse

00

Premium Responders

High spending, not that interested in deals, prefers catalogue and have high acceptance and conversion rate towards campaigns.

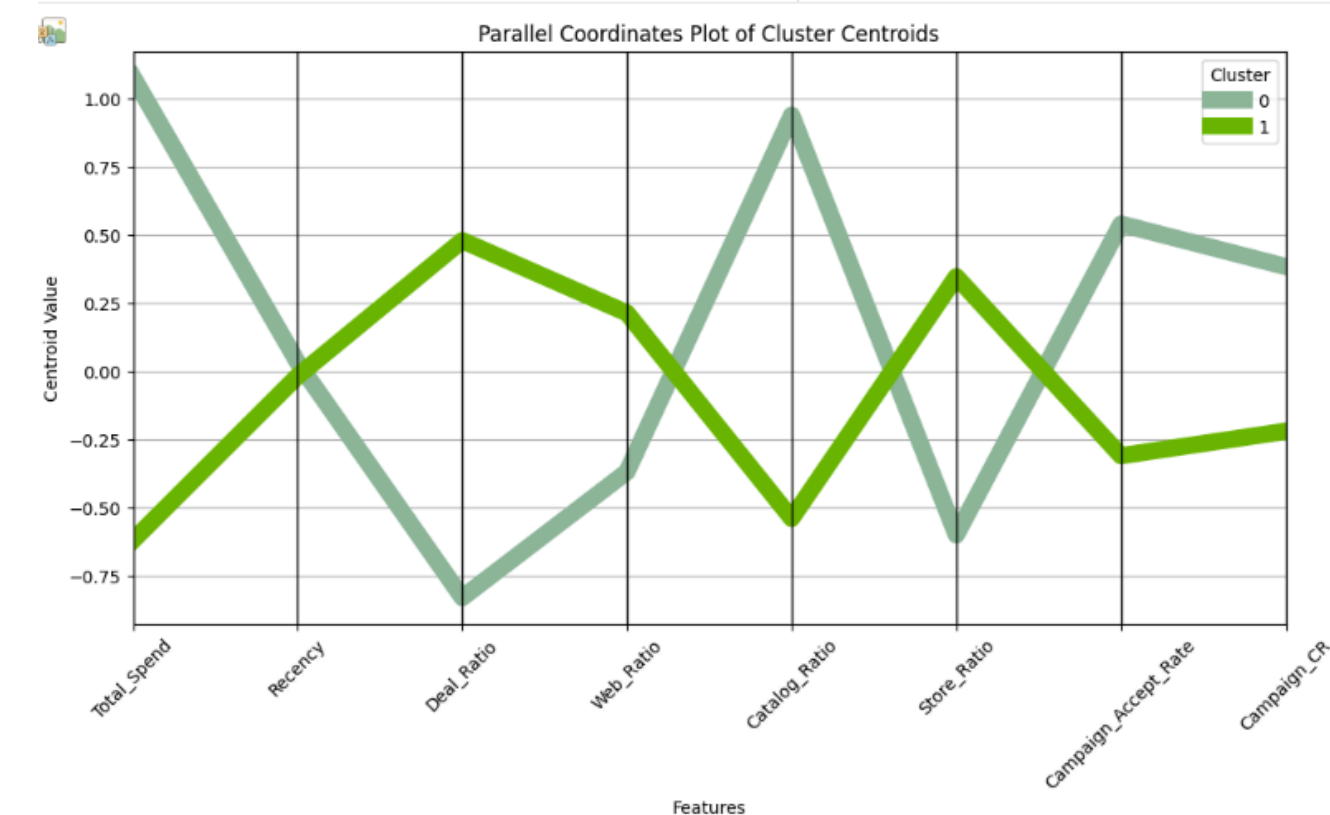
01

Low-Value Deal Hunters

Lower total spending, interested in deals, prefer web and store channels, lower acceptance and conversion rate towards campaign.

Behavior

Clustering



```
behavior_summary = df.groupby('Behavior_Cluster')[feats_behavior].mean()  
display(behavior_summary)
```

	Total_Spend	Recency	Deal_Ratio	Web_Ratio	Catalog_Ratio	Store_Ratio	Campaign_Accept_Rate	Campaign_CR
Behavior_Cluster								
0	2.355686e+05	48.426439	0.322218	0.354800	0.089256	0.555944	0.017626	0.012234
1	1.258276e+06	50.041045	0.096883	0.283824	0.295795	0.412918	0.133333	0.098460

Next steps: [Generate code with behavior_summary](#) [New interactive sheet](#)

Affluent High Value Loyalist -P0B0- (Mature, more educated, high spender, low deals ratio, high campaign acc & conversion) - VIP/High LTV cust - Retain & maximize lifetime value - Reward them, don't discount them	Core Family Loyalists -P1B0- (Avg Aged Cust, medium income, high spender, low deals ratio, high campaign acc & conversion) - Stable core base - Build habit and stickiness using campaign - Maintain consistency & repeat purchases	Emerging High-Potential Families -P2B0- (Young Cust, lower income, high spender, low deals ratio, high campaign acc & conversion) - Growth segment - In-store promotions - Grow them into high-value segment - Invest early → grow long-term value
Affluent Value Seekers -P0B1- (Mature, more educated, high spender, high deals ratio, low campaign acc & conversion) - Wealthy but promo sensitive - Increase margin while keeping engagement - Give value perception, not heavy discounts	Mainstream Deal Shoppers -P1B1- (Avg Aged Cust, medium income, high spender, high deals ratio, low campaign acc & conversion) - Volume segment - Drive volume and frequency - High volume, promo-driven conversion	Early-Stage Bargain Seekers -P2B1- (Young Cust, lower income, low spender, high deals ratio, low campaign acc & conversion) - Acquisition segment - Acquire & increase engagement - Convert → retain → gradually upgrade



THANK YOU

● FOR YOUR NICE ATTENTION

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